

Semester IV							
Sl. No.	Type of course	Code	Course Title	Hours per week			Credits
				L	T	P	
Theory							
1	Professional Core Course	PCC-CSE 401	Database Management System	3	0	0	3
2	Professional Core Course	PCC-CSE 402	Object Oriented Programming	3	0	0	3
3	Professional Core Course	PCC-CSE 403	Machine Learning	3	0	0	3
4	Professional Core Course	PCC-CSE 404	Discrete Mathematics	3	1	0	4
3	Professional Core Course	PCC-CSE 405	Formal Language & Automata Theory	3	0	0	3
6	Humanities & Social Sciences including Management course	HSMC-CSE 401	Introduction to Industrial Management (Humanities-III)	3	0	0	3
Practical							
7	Engineering Science Course	PCC-CSE 491	Database Management System lab	0	0	4	2
8	Professional Core Course	PCC-CSE 492	Object Oriented Programming Lab	0	0	4	2
9	Professional Core Course	PCC-CSE 493	Machine Learning Lab	0	0	4	2
Total credit of the Semester							25

**Detailed Syllabus for B. Tech in Computer Science & Engineering
For Academic Year 2024-2025**



CSE

Second Year - Fourth Semester

Database Management Systems

Code: PCC-CSE 401

Contact: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	To learn data models, conceptualize and depict a database system using ER diagram
2.	To understand the concept of Database Design in Normalization techniques
3.	To understand the internal storage structures in a physical DB design
4.	To know the fundamental concepts of transaction processing techniques
5.	To know the manipulation of SQL Queries, relational algebra queries.

Pre-Requisite:	
1.	Proper understanding of data structures and algorithms
2.	Understanding of set theory
3.	Basic programming knowledge

Unit	Content	Hrs.
1.	Database system architecture: Introduction to DBMS, File based system versus Database System, 3 schema architecture, Data Abstraction, Data Independence, Administration roles, types of users.	4
2.	Data models: Relational model- definition and Properties, Key concepts, integrity constraints. Codd's Rules. ER Model to Relational model. (ER Diagram –Strong entity, weak entity, binary, ternary relationship, cardinality, participation, extended ER feature)	6
3.	Query languages: Relational algebra (operations, Codd's algebraic rules), Evaluation of relational algebra expressions, Querying by non-procedural language (SQL)	5
4.	Relational database design: Single valued functional dependency, Armstrong's axioms, Normal forms (up to BCNF), Properties of decomposition. Multivalued Functional dependency, 4NF. (Overview), , Join dependency and 5NF.	9
5.	Storage strategies: Storage structures (Sequential, indexed sequential) Types of single level index (Primary, secondary, clustering), multi-level index, Dynamic multilevel indices using B-trees and B+ tree (Creation, insertion, deletion)	6
6.	Transaction processing: Concurrency control, ACID property, Serializability of scheduling (conflict and view), Locking (2PL) and timestamp-based schedulers, Database recovery techniques.	6

Text book and Reference books:

1. “Database System Concepts”, 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill.
2. “Principles of Database and Knowledge – Base Systems”, Vol 1 by J. D. Ullman, Computer Science Press.
3. Database Management Systems, R.P. Mahapatra, Khanna Publishing House, New Delhi (AICTE Recommended Textbook – 2018)
4. “Fundamentals of Database Systems”, 5th Edition by R. Elmasri and S. Navathe,
5. Pearson Education “Foundations of Databases”, Reprint by Serge Abiteboul, Richard Hull, Victor Vianu, Addison-Wesley

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 401.1	Understanding Database system architecture and data models.
PCC-CSE 401.2	Design the database with normalization techniques and construct queries using Relational Algebra, SQL etc.
PCC-CSE 401.3	Develop efficient storage scheme of saving and retrieving Records and Files
PCC-CSE 401.4	Understand transaction processing, concurrency control and recovery techniques

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	1	2	1	-	-	-	-	-	-	3	1	1
CO2	2	1	2	2	1	2	1	-	-	-	1	-	3	2	1
CO3	3	2	3	1	2	2	-	2	-	-	-	-	2	2	3
CO4	2	3	3	2	-	2	2	2	2	1	-	-	3	3	1
AVG.	2.25	1.75	2.25	1.5	1.66	1.75	1.5	2	2	1	1	-	2.75	2	1.5

Object Oriented Programming

Code: PCC-CSE 402

Contacts: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	Introduce the principles of object-oriented programming in a higher-level programming language.
2.	Analyze a problem statement to develop a mental model of objects necessary to create software architecture.
3.	Utilize object-oriented programming to frame software architectures, with care towards separation of concerns and abstraction.
4.	Gain skills in designing, and programming software for reuse of code.
5.	Establish development methods in object-oriented programming to qualify students for teaching the language in other settings.

Pre-Requisite:	
1.	Programming Fundamentals

Unit	Content	Hrs.
1.	OOP concepts and Overview of Java language Class, object, Abstraction, encapsulation, Inheritance, Encapsulation Polymorphism, Difference between OOP and Procedural approach. Basics of Java programming, Data types, Variables, Operators, Control structures	8

	including selection, Looping, Java methods, Overloading, Math class, Arrays in java, Basic I/O operations using BufferedReader and Scanner class.	
2.	Objects and Classes Basics of objects and classes in java, Constructors, Finalizer, Visibility modifiers, Methods and objects, Inbuilt classes like String, Character, StringBuffer, StringBuilder, this reference.	8
3.	Inheritance, Polymorphism and Package: Inheritance in java, Super and sub class, Overriding, Object class, Polymorphism, Dynamic binding,, instanceof operator, Abstract class, Interface in java. Creation of package, detailed discussion on access specifiers. Defining packages, Member access, Importing packages. Creation of packages etc. UTIL package.	8
4.	Exception Handling and Multithreading Exception Handling: Exception handling basics, different types of Exception classes, use of try & catch with throw, Errors in Java Program, Exceptions, throws Clause, throw Clause, Types of Exceptions, creation of user defined exception classes. Threads: Single Tasking, Multi-Tasking, thread life cycle, Uses of Threads, creating a Thread and Running it, terminating the Thread, Thread Class Methods, suspending & resuming threads, thread priorities, thread synchronization.	7
5.	Event and GUI programming: Event handling in java, Event types, Mouse and key events, GUI Basics, Panels, Frames, Layout Managers: Flow Layout, Border Layout, Grid Layout, GUI components like Buttons, Check Boxes, Radio Buttons, Labels, Text Fields, Text Areas, Combo Boxes, Lists, Scroll Bars, Sliders, Windows, Menus, Dialog Box, Introduction to swing. Overview of Collections : ArrayList, LinkedList, Vector etc.	6

Text book and Reference books:

1. Rambaugh, James Michael, Blaha – "Object Oriented Modelling and Design" – Prentice Hall, India
2. Ali Bahrami – "Object Oriented System Development" – Mc Graw Hill
3. Patrick Naughton, Herbert Schildt – "The complete reference-Java2" – TMH
4. R.K Das – "Core Java for Beginners" – VIKAS PUBLISHING
5. Deitel and Deitel – "Java How to Program" – 6th Ed. – Pearson
6. Ivor Horton's Beginning Java 2 SDK – Wrox
7. E. Balagurusamy – " Programming With Java: A Primer" – 3rd Ed. – TMH

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 402.1	Apply the concepts of classes and objects to solve simple problems.
PCC-CSE 402.2	Analyze programs using inheritance, packages and interfaces.
PCC-CSE 402.3	Evaluate the use of exception handling mechanisms and a multithreaded model to solve real-world problems.
PCC-CSE 402.4	Design and develop GUI programs.

Mapping of COs with POs and PSOs (Course Articulation Matrix):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	1	-	-	-	1	1	-	1	3	2	2
CO2	2	2	3	3	2	-	1	-	2	-	-	1	2	2	2
CO3	2	2	2	2	3	1	1	-	2	-	-	1	2	2	2
CO4	2	1	3	2	2	1	1	2	2	1	2	3	2	2	3
AVG.	2.00	1.75	2.5	2.25	2.00	0.50	0.75	0.50	1.75	0.50	0.50	1.50	2.25	2.00	2.25

Machine Learning

Code: PCC-CSE 403

Contact: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:

1.	To learn the concept of how to learn patterns and concepts from data without being explicitly programmed
2.	To design and analyses various machine learning algorithms and techniques with a modern outlook focusing on recent advances.
3.	Explore supervised and unsupervised learning paradigms of machine learning.
4.	To explore Deep learning technique and various feature extraction strategies.

Pre-Requisite:

1.	Basic understanding of Linear algebra, Trigonometry, Statistics, calculus, probability
2.	Python Programming

Unit	Content	Hrs.
1.	Some performance metrics associated with ML Supervised Learning (Regression/Classification) Basic methods: Distance-based methods, Nearest-Neighbor's, Decision Trees, Naive Bayes Linear models: Linear Regression, Logistic Regression Support Vector Machines, Nonlinearity and Kernel Methods	12
2.	Unsupervised Learning Clustering: K-means/Kernel K-means / k-medoids Dimensionality Reduction: PCA and kernel PCA Matrix Factorization and Matrix Completion	9
3.	Evaluating Machine Learning algorithms and Model Selection, Introduction to Statistical Learning Theory, Ensemble Methods (Boosting, Bagging, Random Forests)	6
4.	Time-Series Data, basics of Semi-supervised Learning, Reinforcement Learning, DL fundamentals: Introduction to neural network and feed forward network	9

References:

1. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009 (freely available online)
2. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007
3. Dr. Rajiv Chopra, Machine Learning, Khanna Publishing House, 2018
4. Tom M. Mitchell, Machine Learning, McGraw-Hill, 2017
5. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012

Course Outcomes:

On completion of the course students will be able to	
PCC-CSE 403.1	To understand principles and concepts of various ML algorithms
PCC-CSE 403.2	To evaluate the ML algorithms using real life data
PCC-CSE 403.3	To analyse the ML algorithms according to different performance metrics
PCC-CSE 403.4	To understand Ensemble Methods, time series data, semi supervised learning, Reinforcement Learning, and essentials of NN architectures

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	-	-	-	-	-	3	3	3	1
CO2	3	2	2	2	1	-	-	-	-	1	-	3	3	3	2
CO3	3	3	2	3	1	-	-	-	-	1	-	3	3	3	3
CO4	2	3	3	3	-	-	-	-	1	2	-	3	3	3	3
AVG.	2.83	2.83	2.17	2.50	1.00	1.00	1.00	0.00	1.00	1.33	0	3.00	3.00	3.00	2.50

Discrete Mathematics

Code: PCC-CSE 404

Contacts: 3L+1T

Theory: 4 hrs. / Week

Credit points: 4

Course Objective:	
1.	Use mathematically correct terminology and notation.
2.	Construct correct direct and indirect proofs.
3.	To know Syntax, Semantics, Validity and Satisfiability, Graphs and Trees.
4.	Use counterexamples. Apply logical reasoning to solve a variety of problems.

Pre-Requisite:	
1.	Some concepts from basic math – algebra, geometry

Unit	Content	Hrs.
1	Sets, Relation and Function: Operations and Laws of Sets, Cartesian Products, Binary Relation, Equivalence Relation, Partial Ordering Relation, Hasse diagram, Functions, Domain and range of function Inverse and Composite Function, Bijective function, Inverse and Composite Function, Principles of Mathematical Induction, Well-ordering principle, Recursive definition, Division algorithm, Prime Numbers, Greatest Common Divisor: Euclidean Algorithm, Fundamental theorem of arithmetic, Least common multiple, Diophantine equation, Congruence, Properties of congruence, residue classes, Solution of linear congruence, Fermat's theorem, Wilson's theorem.	12
2	Basic counting techniques: Permutation and combination, Principle of inclusion and exclusion, Pigeon-hole principle.	4
3	Propositional Logic: Syntax, Semantics, Validity and Satisfiability, Basic Connectives and Truth Tables, Logical Equivalence: The Laws of Logic, Logical Implication, Rules of Inference, The use of Quantifiers. Proof Techniques: Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency.	8
4	Algebraic Structures and Morphism: Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Sub-group, Quotient Structures, Cyclic Groups, Coset, Normal Subgroups, Permutation Groups, Quotient group. Morphism: Homomorphism and Isomorphism, Kernel and Image. Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields.	10
5	Graphs and Trees: Graph coloring, Chromatic number, Chromatic polynomial, Decomposition theorem, Planarity of a graph, Clique, Clique number, Independent set, Independence number, Brooke's theorem, Tree: Definition, types of tree (rooted, binary), properties of trees. Theorems of tree, Prim's and Kruskal's Algorithms to find MST. Weighted trees and prefix codes. Single source shortest path: Dijkstra's Algorithm. Tree traversal algorithms: BFS & DFS. Dual and Planar graphs, detection of planarity, Theorems on planarity. Matching, Maximal and perfect matching, Matching number, SDR of Matching, Hall Marriage theorem.	14

Text book and Reference books:

1. A text book of Discrete Mathematics, Guruprasad Samanta, New Age International.
2. Gary Haggard, John Schlipf and Sue Whitesides, Discrete Mathematics for Computer Science, CENGAGE Learning
3. J.K. Sharma, Discrete Mathematics, Macmillan
4. S. K. Chakraborty and B. K. Sarkar, Discrete Mathematics, OXFORD University Press.
5. Douglas B. West, Introduction to graph Theory, PHI
6. C. L. Liu, Elements of Discrete Mathematics, 2nd Ed., Tata McGraw-Hill, 2000.
7. N. Deo, Graph Theory, Prentice Hall of India, 1974.
8. S. Lipschutz and M. L. Lipson, Schaum's Outline of Theory and Problems of Discrete Mathematics, 2nd Ed., Tata McGraw-Hill, 1999.
9. J. P. Tremblay and R. P. Manohar, Discrete Mathematics with Applications to Computer Science, Tata McGraw-Hill, 1997.
10. N. Chandrasekaran and M. Umaparvathi, Discrete Mathematics, PHI

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 404.1	Remember the concept of set theory, counting techniques, basic algebraic structures, and graphs. Discuss the logic sentence in terms of predicates, quantifiers, and logical connectives.
PCC-CSE 404.2	Derive the solution for a given problem using deductive logic and prove the solution based on logical inference. Evaluate different relations, and algorithms of number theory, logic, counting techniques, and graph theory.
PCC-CSE 404.3	Classify algebraic structure for a given a mathematical problem. Analyze algorithms and logic related to real-life problems.
PCC-CSE 404.4	Apply functions, basic and advanced principles of counting, logical equivalence, algebraic structures, and algorithms to solve complex engineering problems.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	1	-	-	2	3	3	2
CO2	3	3	3	3	2	1	1	-	2	2	2	3	3	3	1
CO3	3	3	3	3	3	2	-	-	1	1	1	3	3	3	2
CO4	3	3	2	3	3	-	-	2	3	1	2	2	3	2	3
AVG.	3.00	3.00	2.75	3.00	2.50	1.50	1.00	2.00	1.75	1.33	1.67	2.50	3.00	2.75	2.00

Formal Language & Automata Theory

Code: PCC-CSE 405

Contacts: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	Be able to know different classes of grammars and formal languages.
2.	Be able to write grammar and construct finite automata for different languages.
3.	Be able to know properties of regular expression and construct pushdown automata for context free grammars.
4.	Be able to understand undesirability and decide on languages that are undesirable.

Pre-Requisite:	
1.	Basics of Set Theory, Graph Theory
2.	Principle of Mathematical Induction, Proof by Contradiction

Unit	Content	Hrs.
1.	Formal Languages and Grammars: Alphabet, strings, languages, grammars, productions and derivation, Chomsky hierarchy of grammars.	6
2.	Finite Automata: deterministic finite automata (DFA) and equivalence with regular expressions, nondeterministic finite automata (NFA), NFA with epsilon, NFA to DFA conversion, equivalence of NFA and DFA, state transition graph, transition table, Mealy and Moore machine, minimization of automata.	6
3.	Regular grammar and Languages: Regular expressions and languages, regular grammars and equivalence with finite automata, properties of regular languages, Arden's theorem, closure properties of regular set, pumping lemma for regular expression.	9
4.	Context-free grammar and pushdown automata: Context-free grammars (CFG) and languages (CFL), derivation, parse trees, ambiguity in CFG, left recursion, left factoring, normal forms, Closure Properties of CFG. Pushdown automata (PDA): Definition, acceptance of strings by PDA, DPDA and NPDA, construction of PDA from CFG.	9
5.	Turing Machines and Undecidability: The basic model for Turing machines (TM), Turing recognizable (recursively enumerable) and Turing-decidable (recursive) languages and their closure properties, variants of Turing machines, nondeterministic TMs and equivalence with deterministic TMs, unrestricted grammars and equivalence with Turing machines, TMs as enumerators. Undecidability: Church-Turing thesis, universal Turing machine, the universal and diagonalization languages, reduction between languages and Rice's theorem, undecidable problems about languages.	7

Text books:

1. John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman, Introduction to Automata Theory, Languages, and Computation, Pearson Education Asia.
2. Harry R. Lewis and Christos H. Papadimitriou, Elements of the Theory of Computation, Pearson Education Asia.
3. Dexter C. Kozen, Automata and Computability, Undergraduate Texts in Computer Science, Springer.

Reference books:

1. Shyamalendu Kandar, Introduction to Automata Theory, Formal Languages and Computation, PEARSON.
2. Dr. R. B. Patel, Theory of Computation, Khanna Publishing House.
3. K.L.P. Mishra (Author), N. Chandrashekar (Author), Theory of Computer Science: Automata, Languages and Computation, Prentice-Hall of India Pvt.Ltd.

Course Outcomes:

On completion of the course students will be able to

PCC-CSE 405.1	Understand the basics of formal language, grammar, and automata.
PCC-CSE 405.2	Apply pumping lemma to disprove the language is regular language or context free language.
PCC-CSE 405.3	Remember context free grammar to design pushdown automata.
PCC-CSE 405.4	Analyze computational problems and understand decidability.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	3	-	-	1	1	1	1	2	-	2	3	1	2
CO2	2	3	3	2	2	1	1	1	2	2	2	2	3	3	2
CO3	3	3	3	2	2	2	-	-	2	2	1	-	2	3	2
CO4	2	3	3	3	1	2	2	2	3	3	2	2	3	3	2
AVG.	2.00	3.0	3.0	2.33	1.66	1.50	1.33	1.33	2.00	2.25	1.66	2.00	2.75	2.50	2.00

Introduction to Industrial Management (Humanities-III)

Code: HSMC-CSE 401

Contacts: 3L

Theory: 3 hrs. / Week

Credit Points: 3

Course Objective:	
1.	Gain a comprehensive understanding of industrial systems, management principles, and key concepts such as production planning, quality management, and supply chain operations.
2.	Develop practical skills in operations management, leadership, decision-making, and communication to effectively manage industrial processes, teams, and resources.
3.	Stay updated on emerging trends, technologies, and best practices in industrial management to adapt, innovate, and lead change initiatives in dynamic industrial environments.

Pre-Requisite:	
1.	N/A

Unit	Content	Hrs.
1.	Introduction to Industrial Management: Industrial Management –Concept Meaning and Definitions, Scope of Industrial Management, School of management thoughts – H. Fayol & F.W.Taylor	4
2.	Production, Planning and Control: Concept and Meaning of PPC , Job shop scheduling	4
3.	Plant Location, Plant Layout and Work Study: Product design process, Process selection, Types of production system (Job, Batch and Mass Production), Concept and Meaning of Plant Location, Theories of Plant Location, Plant location-factors-Urban & Rural sites comparison, Concept and Meaning of Plant Layout, Factors Affecting Plant Layout, Advantages of Plant Layout, Techniques of Plant Layout, Types of plant layouts, Work study.	6
4.	Statistical Quality Control: Variables - attributes, Shewart control chart for variables - X chart, R chart, - Attributes-Defective-Defect Charts for attributes-p chart, c chart (simple problems).	4
5.	Material Management and Purchase Management: A) Material management-definition, functions, importance, relationship with other departments. B) Purchase - objectives, purchasing systems, purchase procedure, terms and forms used in purchase department. C) Storekeeping- functions, classification of stores as centralized and decentralized with their advantages, disadvantages and application in actual practice. D) Inventory control: i. Definition. ii. Objectives. iii. Economic Order Quantity (EOQ) and numeric examples. iv. ABC analysis and other modern methods of analysis. E) Material Requirement Planning (MRP) concept, applications and brief details about software packages available in market.	8
6.	Recent trends in Industrial Management: A) ERP (Enterprise resource planning) - concept, features and applications. B) Important features of MS Project. C) Logistics- concept, need and benefits. D) Just in Time (JIT)-concept and benefits. E) Supply chain management-concept and benefits.	4

Text book and Reference books:

1. O. P. Khanna – “Industrial Engineering & Management”, Dhanpat Rai Publications
2. S.C. Sharma - “Engineering Management – Industrial Engineering & Management”, Khanna Book Publishing Company, New Delhi
3. A.P. Verma and K.L. Maheshwari – “Industrial Engineering and Management”, Sultan Chand & Sons
4. N. Nair – “Materials Management”, Tata McGraw Hill Education

Course Outcomes:	
On completion of the course students will be able to	
HSMC-CSE 401.1	Students will be able to understand and recall the fundamental concepts, principles, and theories of industrial management.
HSMC-CSE 401.2	Students will be able to apply different industrial management tools and techniques by utilizing software and implementing quality management strategies.
HSMC-CSE 401.3	Students will be able to evaluate the benefits and weaknesses of different industrial management strategies and approaches as well as they will be able to assess the market trends.
HSMC-CSE 401.4	Students will be able to assess whether the strategies are effective in the line of organizational goal and create innovative solutions to face the market challenges

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	2	1	1	3	2	2	2	2	2	2	2	1	2
CO2	2	1	3	2	2	1	1	1	1	1	2	3	2	3	3
CO3	1	2	3	2	1	2	2	1	1	2	2	2	2	3	3
CO4	1	2	2	3	2	1	1	1	1	2	2	2	1	2	3
AVG.	1.25	1.5	2.5	2.00	1.50	1.75	1.50	1.25	1.25	1.75	2.00	2.25	1.75	2.25	2.75

Database Management System Lab

Code: PCC-CSE 491

Contacts: 4P

Practical: 4 hrs. / Week

Credit Points: 2

Course Objective:	
1.	To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modelling, designing, and implementing a DBMS.
2.	To construct the SQL queries for a given specification in Open source and Commercial DBMS – ORACLE platform.
3.	To enable the students to practice the concepts learnt in the subject Database Management System by developing a database.

Pre-Requisite:	
1.	Proper understanding of data structures and algorithms
2.	Understanding of set theory
3.	Basic programming knowledge

Laboratory Experiments:	
Structured Query Language	No. of sessions
1. Creating Database Creating a Database Creating a Table Specifying Relational Data Types Creating Constraints	1
2. Table and Record Handling INSERT, DELETE, UPDATE, TRUNCATE commands DROP, ALTER commands COMMIT, ROLLBACK commands	1
3. Retrieving Data from a Database	4

	The SELECT statement Using the WHERE clause ORDER BY, GROUP BY and HAVING Clause Using Logical Operators, Using IN, BETWEEN, LIKE, IS operators Using ANY, ALL, EXISTS operator Using Aggregate Functions Combining Tables Using JOINS Subqueries(independent and correlated)	
4.	Database Management Creating Views Creating Column Aliases Creating Database Users Using GRANT and REVOKE	2
5.	Introduction to PL/SQL Basic programming constructs (decision making, iteration) Practical implementation of Stored Procedure and function Cursor Trigger	4

(Detailed instructions for Laboratory Manual to be followed for further guidance)

Resources

1. Sql, pl/sql the programming language of oracle by Bayross, Ivan
2. <https://livesql.oracle.com/>

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 491.1	Create a database schema, relation and implement referential integrity constraint.
PCC-CSE 491.2	Implement and control the database using SQL queries for a given specification in ORACLE platform
PCC-CSE 491.3	Applying administrative activities like working with views, creating users and granting or revoking privileges on database objects.
PCC-CSE 491.4	Create database objects by using Procedural extension of SQL to develop a better control on database.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	-	-	1	1	-	-	-	-	-	1	-	-
CO2	2	2	3	1	1	1	1	-	-	-	-	-	2	1	-
CO3	3	3	3	3	2	2	-	2	2	2	2	1	3	3	3
CO4	2	3	3	2	-	1	-	-	-	-	1	-	3	3	3
AVG.	2.25	2.25	2.5	2	1.5	1.25	1	2	2	2	1.5	1	2.25	2.33	3

Object Oriented Programming Lab

Code: PCC-CSE 492

Contacts: 4P

Practical: 4 hrs. / Week

Credit Points: 2

Course Objective:	
1.	To build software development skills using java programming for real-world applications.
2.	To understand and apply the concepts of classes, packages, interfaces, inheritance,
3.	Exception handling and file processing.

Pre-Requisite:	
1.	Programming Fundamentals

Laboratory Experiments:	
1.	Assignments on basic programming. Decision control, Loop Control, Array.
2.	Assignments on class, constructor, overloading, inheritance, overriding.
3.	Assignments on Strings.
4.	Assignments on developing abstract class interfaces- multiple inheritance, extending interfaces
5.	Assignments on creating and accessing packages
6.	Assignments on multithreaded programming, Exception handling programming.
7.	Assignments on AWT and Event handling programming, collections.
Note: Use Java for programming	

(Detailed instructions for Laboratory Manual to be followed for further guidance)

1. Timothy Budd, Understanding object-oriented programming with Java, 2nd ed., Pearson, 2001
 2. Herbert Schildt, The complete reference Java 2, TMH, 8th ed., 2011.
 3. Beginning Programming with Java for Dummies, 5ed by Barry A. Burd
- (Detailed instructions for Laboratory Manual to be followed for further guidance)

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 492.1	Understand java programs using object oriented programming concepts.
PCC-CSE 492.2	Apply the knowledge of inheritance, interface to develop Java programs.
PCC-CSE 492.3	Implement and evaluate simple applications using object oriented concepts such as package, exceptions.
PCC-CSE 492.4	Create GUIs and event driven programming applications for real world problems.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	2	1	-	-	-	1	1	-	2	2	3	2
CO2	2	3	3	2	2	-	-	-	1	1	-	2	3	2	2
CO3	2	3	3	2	3	1	1	-	1	2	1	2	2	3	2
CO4	3	3	3	2	3	2	1	1	2	2	2	3	3	3	3
AVG.	2.25	3.00	2.75	2.00	2.25	0.75	0.50	0.25	1.25	1.50	0.75	2.25	2.50	2.75	2.25

Machine Learning Lab

Code: PCC-CSE 493

Contacts: 4P

Practical: 4 hrs. / Week

Credit points: 2

Course Objective:	
1.	To understand the basic theory underlying machine learning.
2.	To be able to formulate machine learning problems corresponding to different applications.
3.	To understand a range of machine learning algorithms along with their strengths and weaknesses.
4.	To be able to apply machine learning algorithms to solve problems of moderate complexity.
5.	To apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.

Pre-Requisite:	
1.	Statistics, Probability, Linear algebra, calculus and
2.	Programming language (Python)

Laboratory Experiments:	
1.	Study some python libraries associated with ML
2.	Write a program in python to implement linear regression in real life dataset
3.	Write a program in python to implement logistic regression in real life dataset
4.	Write a program in python to implement k-nearest neighbour algorithm to classify a real-life dataset.
5.	Write a program in python to implement Naive based algorithm in real life dataset classification.
6.	Write a program in python to demonstrate the working of decision-tree algorithm. Use an appropriate dataset for building the decision trees and apply this knowledge to classify a new sample.
5.	Write a program to demonstrate the working of SVM algorithm. Use an appropriate dataset and apply this knowledge to classify the new sample.
6.	Write a program to implement random-forest algorithm to classify real life dataset.
7.	Write a program to implement k-means algorithm to classify real life dataset.
8.	Write a program to implement principal component analysis (PCA) in real life dataset.
9.	Write a program to implement random-forest algorithm to classify real life dataset.
10.	Write a program to implement ANN algorithm to classify real life dataset.

(Detailed instructions for Laboratory Manual to be followed for further guidance)

Text book and Reference books:

1. Manaranjan Pradhan and U Dinesh Kumar, machine Learning using Python, First edition, Wiley India Pvt. Ltd, 2019.
2. Andreas C.Mueller, Sarah Guido, Introduction to Machine Learning with Python guide for data scientists, O'REILLY Media, 2016.
3. Abhishek Vijayvargia, Machine Learning with Python, BPB Publications, 2018
4. Practical handbook of ML by S. Bhattacharya & S. Bhattacharya – G. K. Publication

Course Outcomes:	
On completion of the course students will be able to	
PCC-CSE 493.1	Understand different machine learning techniques.
PCC-CSE 493.2	Apply the knowledge of machine learning in various practical fields.
PCC-CSE 493.3	Analyse the performances of different machine learning algorithms.
PCC-CSE 493.4	Develop machine learning models to solve real life problems.

Mapping of COs with POs and PSOs (Course Articulation Matrix):															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	1	-	1	3	3	3
CO2	3	3	3	3	-	-	-	-	1	-	-	2	3	3	3
CO3	3	3	3	3	2	-	-	-	1	-	-	2	3	3	3
CO4	3	3	3	3	3	-	-	-	1	1	-	2	3	3	3
AVG.	3.00	3.00	3.00	3.00	2.75	0.00	0.00	0.00	1.00	1.00	0.00	1.67	3.00	3.00	3.00